Attention-Driven Cross-Modal Remote Sensing Image Retrieval

Re-implementation and adaptation to EuroSAT RGB and SAR imagery

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1. Introduction and objective

Satellites observe the same scene with different sensors (optical RGB, multispectral, SAR, ...). Cross-modal retrieval takes a query from one modality (M1) and returns semantically relevant images (same land-use class) from a gallery of another modality (M2). The assignment asks us to implement an assigned research paper and reproduce its results. This report documents our implementation of *Attention-Driven Cross-Modal Remote Sensing Image Retrieval* (Chaudhuri, Banerjee, Bhattacharya, Datcu, IGARSS 2021), adapted to the datasets named in the assignment sheet: EuroSAT RGB and EuroSAT-SAR. The retrieval directions evaluated are RGB to SAR and SAR to RGB, measured by mean average precision (mAP).

2. Original paper and methodology

The paper (Att-CMIR-Net) extends CMIR-Net. Two ImageNet-pretrained VGG-16 encoders process the two modalities (photo and hand-drawn sketch). The last convolutional map of the *sketch* encoder is globally average-pooled, passed through a fully connected layer and a sigmoid; the resulting vector multiplies the fully connected features of the *photo* encoder (attention). Both streams are mapped to a shared 128-d space trained with four losses: $L_1 = ||V_a - V_b||^2$ (alignment), $L_2 = CE(V_{ab})$ (classification), $L_3 = ||V_a||^2 + ||V_b||^2$ (norm) and $L_4 = ||w_{ab}V_a - V_b||^2 + ||w_{ba}V_b - V_a||^2$ (cross-decoders); $L = L_1+L_2+L_3+L_4$, optimised with SGD (lr 0.001, batch 128, about 40-50 epochs). Retrieval ranks the gallery by Euclidean distance. The paper evaluates on Earth on Canvas (14 classes, 100 photo and 100 sketch images per class, 70:30 split).

The values below are the paper's **published** results on a different dataset and task (photo/sketch). They are quoted for context only and are not results of this project.

Direction (published, Earth on Canvas)	mAP	P@10
Sketch to Photo	0.753	0.784
Photo to Sketch	0.723	0.745
Sketch to Sketch	0.775	0.788
Photo to Photo	0.804	0.823

3. Dataset and preprocessing

Data. EuroSAT (Sentinel-2 RGB, 64x64 px, 10 land-use classes, 27,000 patches) and EuroSAT-SAR (27,000 dual-polarisation Sentinel-1 patches, GeoTIFF, 2 channels). **Pairing.** The EuroSAT-SAR description states that each Sentinel-2 patch was matched with one Sentinel-1 patch using geospatial coordinates and that the folder structure mirrors EuroSAT. We pair samples by (class folder, filename stem); a check on our copy found all 27,000 stems present in both modalities. The documentation does not state pixel-level alignment or same-day acquisition, so we treat the pairs as geographically matched, not as pixel-registered image pairs (inspect_data.py provides a weak empirical check).

Split. Stratified by class, 70% / 15% / 15% (train / validation / test), seed 42, performed over *pairs* so both modalities of a patch always fall in the same split (splits/*.csv). Test pairs are never used for training, SAR normalisation statistics or checkpoint selection. EuroSAT patches come from many locations but neighbouring patches can be similar, so residual leakage between splits cannot be fully excluded from metadata alone. **Preprocessing.** RGB: scaled to [0,1], ImageNet mean/std. SAR: non-finite values replaced by the channel median; per-channel 1st-99th percentile clipping and standardisation, statistics fitted on the training split only and stored in the checkpoint. Both inputs are bilinearly upsampled from 64 to 128 px. Training augmentation: random flips and transpose.

4. Model architecture

Two VGG-16 convolutional encoders (RGB: 3 input channels; SAR: first convolution rebuilt for 2 channels by averaging the ImageNet filters over RGB and tiling them). Each is followed by global average pooling and a 512-unit FC+ReLU+dropout block. SAR is the attention source (the structural modality, analogous to the sketch). The attention vector sigmoid(FC(GAP(SAR conv map))) multiplies the RGB FC features. A shared MLP (512-256-128) produces the 128-d embeddings V_a (RGB) and V_b (SAR); a shared linear classifier, and two linear cross-decoders implement L_2 and L_4.

Aspect	Paper	This implementation
Modalities	Photo + sketch (Earth on Canvas)	RGB + dual-pol SAR (EuroSAT), as listed in the assignment
Attention source	Sketch encoder	SAR encoder (an analogy; SAR is not a sketch)
Input / feature map	224 px assumed (7x7x512)	64 px upsampled to 128 px (4x4x512)
Batch size	128	16 (6 GB GPU), mixed precision
Epochs	about 40-50	20 (best epoch 18, chosen on validation mAP)
Loss reduction	Frobenius norm (sum)	Mean over batch and dimensions by default (configurable)
Inference	Not specified for unpaired queries	Each modality embedded independently (pair-free RGB path)
Extra training terms	-	Auxiliary CE + distillation of the pair-free RGB path towards the attended embedding
Unspecified details	FC sizes, V_{ab}, decoders, pairing	Our assumptions: MLP sizes, shared classifier, linear decoders
Training stage	Single stage (this paper)	Single stage, end-to-end

5. Training configuration

Setting	Value
Seed (split, init, shuffling)	42
Optimiser	SGD, lr 0.001, momentum 0.9, weight decay 0.0005
Epochs / batch size	20 run (configured 20) / 16
Input size / embedding	128 px / 128-d
Attention	True
Pairing mode	instance
Mixed precision	True
Loss weights (w1..w4, aux)	1.0, 1.0, 1.0, 1.0, 1.0 (mean)
Split sizes (train/val/test)	18900 / 4050 / 4050
Best epoch (by validation mAP)	18
Training time	49.9 min
Hardware / software	NVIDIA GeForce RTX 4050 Laptop GPU | torch 2.12.0+cu132 | python 3.11.9

The best validation epoch (18) is close to the last epoch (20), so training may not have fully converged; longer training was not tried.

6. Experimental protocol

Relevance is class-level: a gallery item is relevant to a query iff it has the same land-use label. Gallery and queries are the held-out test pairs; RGB to SAR uses RGB queries and the SAR gallery, and vice versa. Items are ranked by Euclidean distance between 128-d embeddings. AP is computed over the full ranking, mAP is the mean over queries; P@K is precision in the top K. In the *standard* cross-modal variant the query's own paired counterpart remains in the gallery; the *strict* variant removes it. Same-modality retrieval excludes the query itself. Queries with no relevant item would be skipped (none occur with balanced classes). A random-embedding row gives the chance level (about the class prior, 0.1 for 10 balanced classes).

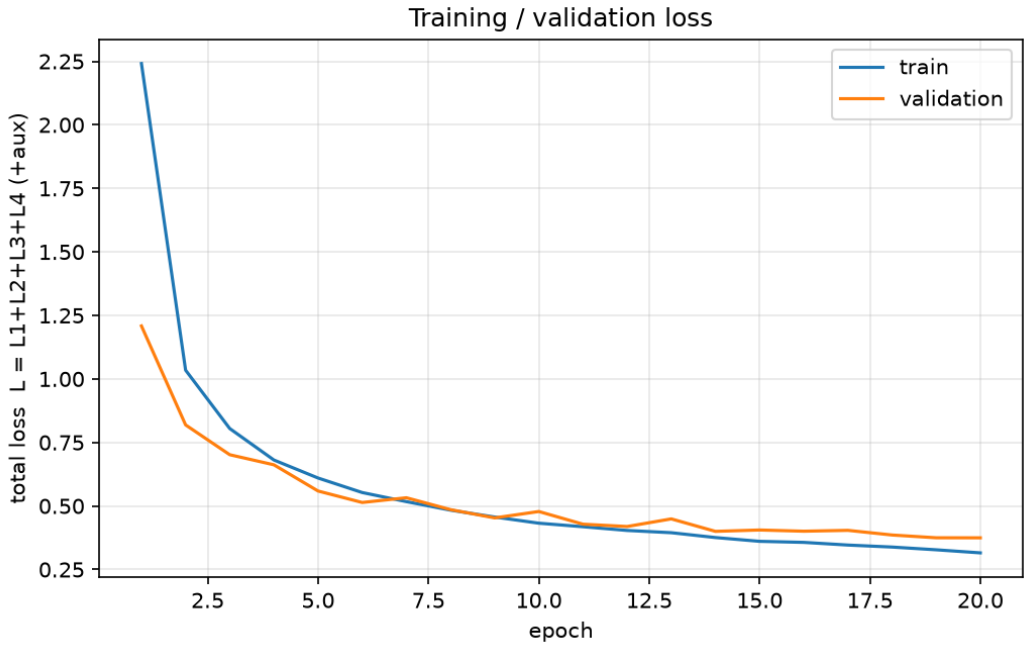
7. Results (measured in this project)

Test split: 4050 pairs; checkpoint epoch 18; seed 42; raw embeddings. Values are fractions in [0, 1].

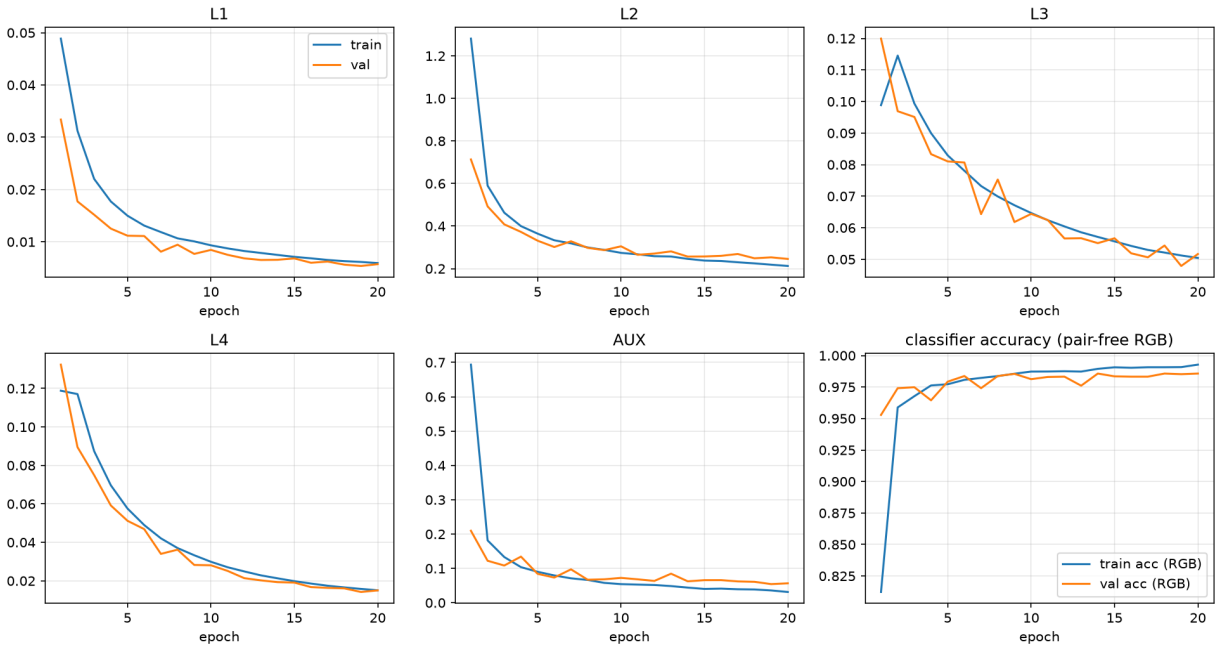
Direction	mAP	P@10	P@20	P@50
RGB to SAR	0.9139	0.9758	0.9750	0.9708
SAR to RGB	0.8643	0.8313	0.8361	0.8435
RGB to SAR (strict)	0.9138	0.9758	0.9749	0.9707
SAR to RGB (strict)	0.8641	0.8309	0.8357	0.8433
RGB to RGB	0.9822	0.9791	0.9799	0.9807
SAR to SAR	0.7930	0.8282	0.8259	0.8227
chance (random embeddings, RGB to SAR)	0.1032	0.0994	0.1004	0.1012

No attention ablation results were found (reports/no_attention/test_results.json).

8. Training and validation loss



Total loss per epoch (train and validation).



Individual loss terms and classifier accuracy.

9. Comparison with the published results

The published numbers (Section 2) are for photo-sketch retrieval on Earth on Canvas with 14 classes (chance level about 0.07). Our experiments use RGB-SAR retrieval on EuroSAT with 10 classes (chance level about 0.10), a different task, dataset, class count and split. The absolute mAP values are therefore **not comparable**, and we do not claim to have reproduced the paper's numbers. What is reproduced is the method (sketch-style cross-attention between two VGG-16 encoders with the four-term loss in a shared 128-d space) and the evaluation measure; the paper's central claim (attention helps relative to a no-attention baseline) is examined by the ablation above, where available.

10. Limitations and conclusion

Limitations. (i) The dataset differs from the paper's; SAR is an analogy for the sketch, not an equivalent. (ii) Several architectural details are not specified in the paper and are our assumptions (Section 4). (iii) At inference, attention cannot use a paired SAR image, so the attention benefit reaches the pair-free path only through training (auxiliary distillation); this is our design, not the paper's. (iv) Loss terms L_1 and L_3 admit a trivial collapse, which only L_2 counteracts; the weights were not tuned. (v) Results come from a single seed per configuration. (vi) The EuroSAT RGB-SAR pairing is geographic according to the dataset documentation; pixel alignment and acquisition date were not verified. (vii) Mixed precision and GPU kernels leave small sources of non-determinism, so re-training gives close but not bit-identical metrics. (viii) The input size and batch size differ from the paper because of the 6 GB GPU and the 64 px EuroSAT patches.

Conclusion. We implemented an attention-driven RGB-SAR cross-modal retrieval network following Att-CMIR-Net, with train/test separation, fixed seeds, saved checkpoints and loss plots, and evaluated both retrieval directions with mAP and P@K on a held-out test split. The measured values are listed in Section 7; interpretation should respect the single-seed and dataset-mismatch caveats above.